Enhancing Internet Traffic Prediction Accuracy Using an Ensemble Approach with a Novel Weight Integration Method and Temporal Convolutional Networks

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***Abstract*—As network technologies evolve, accurate internet traffic prediction has become vital for efficient resource management. This paper introduces a novel ensemble‐learning framework that integrates Temporal Convolutional Networks (TCNs) with a No Negative Constraint Theory (NNCT)-based weighting strategy. Unlike traditional approaches that rely on simple averaging or conventional weight constraints, our method uses the limited-memory Broyden-Fletcher-Goldfarb-Shanno algorithm with box constraints (L-BFGS-B) to optimize weight coefficients precisely within predefined bounds, thereby avoiding the instability often associated with metaheuristic algorithms. Experimental results demonstrate significant improvements in both prediction accuracy and robustness, particularly during periods of high demand. Our approach outperforms current leading forecasting techniques, validating its effectiveness and its potential to enhance network management.**

***Index Terms*—Algorithms on artificial intelligence in networking, internet traffic prediction, traffic analysis, time series forecasting, weight integration.**

I. Introduction and Motivation

With technological advancements, the demand for internet access and related expenses has been rising rapidly [1]. The emergence of 6G and the growing prevalence of the Internet of Things (IoT) [2] have made it crucial to provide an Internet Access Service (IAS) that can efficiently allocate resources. Dynamic management of network resources has become a critical issue among communication scholars [3]. Furthermore, in recent years, various countries have emphasized energy issues and enduring development [4]; thus, reducing the power consumption of network equipment is also an important problem.

Internet traffic prediction is crucial in Software-Defined Networking (SDN) for virtualizing network resources into flexible slices to meet diverse customer requirements. Accurate traffic prediction helps analyze network behavior, forecast demands, and enable dynamic scheduling and efficient resource utilization [5]. Similarly, in 5G/6G and other backbone networks, precise traffic forecasts assist operators in planning, allocating resources, improving efficiency, meeting Quality of Service (QoS) requirements, and reducing operational costs [6].

By 2030, the information and communication technology (ICT) industry could consume up to 20% of global energy [7]. To address environmental concerns, network equipment now integrates renewable energy solutions, like photovoltaics, with intelligent network management systems. These systems use traffic prediction models to guide energy use and reduce reliance on conventional power sources [8].

Therefore, the motivation for developing an accurate traffic prediction method is that it is vital for an intelligent network management system to efficiently allocate resources, reduce power usage, and meet rising internet demand while minimizing the environmental impact of the growing ICT industry.

II. Related Work and Offered Contribution

Internet traffic prediction is a time series prediction problem. In the past, tasks related to time series prediction often used statistical models, such as methods derived from Autoregressive Integrated Moving Average (ARIMA) [9]. In the past decade, machine learning techniques have progressively advanced, with methods such as Support Vector Machines (SVM) [10] and Random Forests (RF) [11] demonstrating notable performance. Nevertheless, many conventional machine learning approaches fall short in traffic forecasting tasks due to the highly nonlinear and nonstationary nature of network traffic [12]. As a result, recent research has increasingly turned to deep learning models, which are better suited to address these complexities.

Within deep learning, convolutional neural networks (CNN), gated recurrent unit (GRU) and long short-term memory networks (LSTM) have been widely used to address time-series forecasting problems, achieving promising results [13], [14], [15]. More recently, the temporal convolutional network (TCN) was proposed [16], which is similar to a 1-D CNN and is particularly designed for time-series forecasting.

Despite the promising models that have been proposed, ensuring their robustness remains a significant challenge. Some studies have applied ensemble learning [17]; however, these methods often lack an empirical basis, such as using average weights, and the weights are typically set to be positive. This

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TABLE I
OUR CORE CONTRIBUTIONS COMPARED TO EXISTING WORKS

Contribution	Our proposed work	Existing works
Improving Accuracy with NNCT-Based Weight Integration	We propose a novel internet traffic prediction method that leverages TCNs, combined with NNCT-based weight integration in ensemble learning, allowing for more flexible weight coefficients. This approach unlocks the potential of ensemble learning, aiming to enhance accuracy and robustness.	Most works tend to overlook robustness in internet traffic forecasting or attempt to use ensemble learning with simple average weights and traditional weight constraints [17], which results in limitations when it comes to improving internet traffic prediction.
Improving Stability and Efficiency with L-BFGS-B	We propose using L-BFGS-B to optimize weight coefficients in NNCT-based weight integration for ensemble learning. This ensures that weight coefficients remain within predefined bounds, enhances stability, and prevents large variations between executions.	Most works in ensemble learning fail to fully unlock their potential, often relying on heuristic algorithms to find optimal weight coefficients, which can result in high variance between executions and lead to instability [20].

approach does not fully unlock the potential for optimization in aggregate. In response to these limitations, we introduce an innovative method that incorporates an empirically grounded weight integration scheme based on the No Negative Constraint Theory (NNCT) [18]. Unlike existing approaches, which often lack flexibility in ensemble design, our method directly addresses this issue and is expected to significantly improve the accuracy of internet traffic forecasting. To the best of our knowledge, this perspective has not been thoroughly explored in previous studies.

To enhance weight optimization in ensemble learning, we adopt the L-BFGS-B algorithm (Limited-memory Broyden-Fletcher-Goldfarb-Shanno with box constraints) [19], which is tailored for problems involving bounded variables, ensuring that all weights adhere to predefined constraints. Unlike various metaheuristic methods [20], L-BFGS-B is grounded in solid theoretical principles and delivers stable performance. Its computational efficiency also enables fast weight determination for short-term forecasts, supporting the timely application of the proposed traffic prediction approach—an element often overlooked in earlier studies.

Based on the above, we propose a novel method that leverages TCNs, combined with the novel L-BFGS-B optimized NNCT-based weight integration in ensemble learning. This method aims to improve the accuracy and robustness of internet traffic prediction. Our core contributions compared to existing works are shown in Table I.

III. PROPOSED METHOD

This section describes the proposed method for internet traffic prediction, including the flowcharts, detailed processes, and its components, such as the forecasting model, the NNCT-based weight integration method, and the optimization algorithm.

A. Proposed Internet Traffic Prediction Method

Figure 1 presents the proposed flowchart for internet traffic prediction. First, the internet traffic data is preprocessed, and important features are extracted from the raw data. The lag value refers to the number of previous hours (delayed hours) that are used as input variables for the models, while the number of estimators (N) indicates how many models are employed in the ensemble learning method. Next, N TCN models

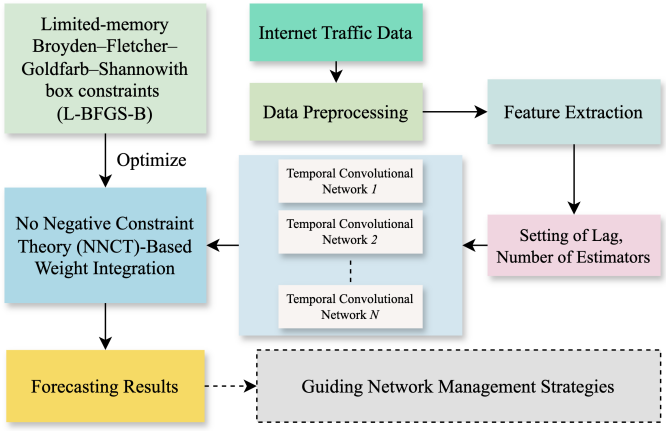


Fig. 1. Flowchart of proposed internet traffic prediction method.

are constructed and utilized for prediction. Afterward, the L-BFGS-B algorithm is used to optimize these weight coefficients in the NNCT-based weight integration prediction method. Finally, the results from the TCN models, with optimized weight coefficients, are used to generate the final forecasting results. Detailed aspects are described in the following sections.

The forecasting results provide valuable insights for network management strategies. This paper aims to develop an accurate and robust method for internet traffic forecasting.

B. Ensemble Learning and No Negative Constraint Theory

Ensemble learning reduces the high error typically seen in individual models by combining the outputs of multiple estimators to enhance overall predictive performance. Through diverse aggregation strategies, it enhances prediction robustness and accuracy while effectively reducing variance.

Given N estimators that produce predictions $f_1(x), f_2(x), \ldots, f_N(x)$ for the same input x, the final output is obtained by integrating these predictions using a weighted combination, as illustrated in Eq. (1).

$$y(x) = \sum_{i=1}^{N} w_i \cdot f_i(x) \tag{1}$$

where w_i represents the weight assigned to the i-th estimator, and $f_i(x)$ denotes its corresponding prediction.

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In ensemble learning, properly adjusting the weight coefficients of individual models is critical to overall performance. While uniform averaging is commonly used, it lacks empirical grounding and restricts weights to positive values only—preventing the assignment of negative weights that could improve predictive accuracy. To overcome this limitation, we introduce the No Negative Constraint Theory (NNCT) as a more flexible and data-driven alternative.

To determine the optimal weight coefficients in NNCT-based ensemble learning [20], the objective is to minimize the total error S_e, formulated as follows [21]

$$S_e = \sum_{t=1}^{T}\sum_{i=1}^{m} w_i e_{it} \tag{2}$$

where e_{it} represents the mean squared error of the i-th estimator at time i. The weight coefficients $(w_1, w_2, \dots, w_N)$ are bounded within the range $[-2, 2]$ under the NNCT framework, thereby lifting the conventional constraint that limits weights to the $[0, 1]$ interval.

The detailed process of the proposed method is presented in Fig. 2, which includes two distinct stages:

Stage 1: We construct a model that mitigates overfitting by monitoring errors from the training and validation sets using an early stopping mechanism. We then apply this model to generate predictions for the training and test sets, and forward these results to Stage 2. This entire stage is repeated for N times.

Stage 2: the L-BFGS-B algorithm is used to generate optimized weights $(w_1, w_2, \dots, w_m)$ for the models using the training set. Then, we apply these weights and the predicted results of the test set for each model to Eq. (1) to generate the final predicted results for the test set.

C. *Limited-Memory Broyden–Fletcher–Goldfarb–Shannowith Box Constraints (L-BFGS-B) Optimization Algorithm*

To minimize the error function S_e defined in Eq. (2), we formulate the following constrained optimization problem:

$$\min_{x\in\mathbb{R}^n} S_e(x)$$
$$subject\ to -2 \le x_i \le 2,\ \forall_i = 1,2,\dots,n \tag{3}$$

Here, $x = [x_1, x_2, \dots, x_n]^T$ is the weight vector for the models. The constraints $-2 \le x_i \le 2$ ensure that each weight x_i remains within the NNCT-based weight integration range.

To efficiently address the bounded-constrained optimization involved in weight tuning, we adopt L-BFGS-B—a quasi-Newton method well-suited for such tasks. Designed to enforce variable bounds while maintaining low memory usage, L-BFGS-B offers high computational efficiency [22], making it particularly suitable for short-term internet traffic prediction. By leveraging gradient information and approximating the Hessian matrix, it delivers a stable and reliable solution, outperforming many metaheuristic approaches in both speed and consistency.

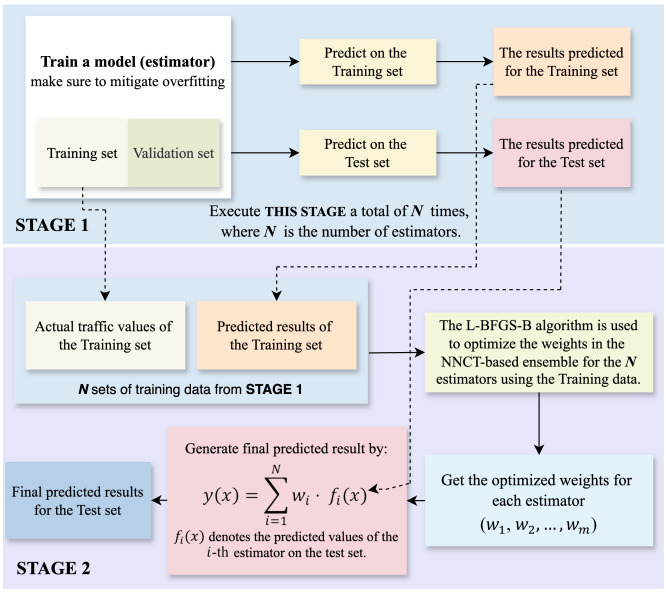


Fig. 2. Detailed two-stage process diagram of the proposed method.

D. *Temporal Convolutional Networks*

Owing to internet traffic prediction being a time-series forecasting task, this paper suggests utilizing the temporal convolutional network (TCN), which is a modified form of 1D-CNN. It has outperformed the recurrent neural network (RNN) architecture in terms of accuracy over a variety of datasets. The network provides two main advantages over 1D-CNN:

1) Long-term memory is provided by residual connection and dilated convolutional layers, which offers it an edge when processing time series [23].

2) Dilated convolutions are used to efficiently manage the complications in backpropagation training as the input data increases and the network becomes deeper, as shown in Fig. 3. They prevent the history size from growing linearly with depth.

Drawing from the work presented in [24], we can define the operation of a dilated convolution on a time series. Consider a sequence X, represented as $(x_0, x_1, \dots, x_t)$, and a filter F, denoted as $(f_1, f_2, \dots, f_k)$. When this filter is applied to the sequence with a dilation factor d, the convolution at any given point x_t in the sequence is expressed as:

$$(F_d^* X)(x_t) = \sum_{k=1}^{K} f_k x_{t-(K-k)d} \tag{4}$$

where d indicates the expansion factor and k denotes the convolution kernel size.

When dealing with a large amount of past data that requires additional layers, the vanishing gradient problem can be mitigated by using a residual block (ResNet). In a standard ResNet, the input x is processed to produce an intermediate output $F(x)$. This output is then added directly to the original input x, resulting in the final output as:

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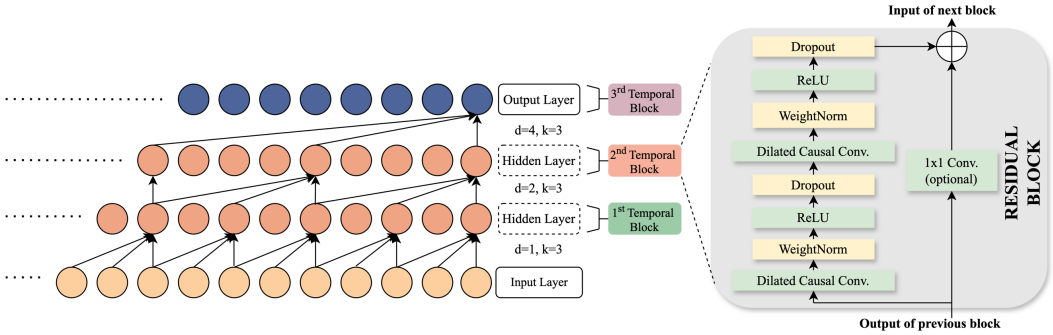


Fig. 3. The network structure of TCN.

TABLE II
DATASET OVERVIEW

Internet Traffic Dataset in State University of Ceará, Brazil [5],[26]	
Period	January 16th, 2019 – July 15th, 2019
Granularity	Hourly data, with bandwidth values aggregated per hour.
Features	Average, maximum, and minimum bandwidth usage (in bps) per hour.

$$H(x) = F(x) + x \tag{5}$$

where $H(x)$ represents the network output, $F(x)$ refers to the denotes the learned residual mapping, and x is the input.

As illustrated in Fig. 3, the residual block of the TCN comprises two layers. Each layer contains a dilated causal convolution, followed by weight normalization, a ReLU activation function, and dropout for regularization purposes.

Although these methods enhance the performance of individual models, they often struggle with the high noise present in time-series internet traffic data, leading to reduced robustness. To address this, we propose combining them with a weight integration strategy based on the NNCT, aiming to improve both the stability and accuracy of internet traffic prediction.

IV. EXPERIMENTS

A. Data Preparation

The dataset overview is described in Table II. It is sequentially divided as follows: the first 75% is allocated to the training set, the next 15% to the validation set, and the final 10% to the test set. Sequential splitting helps prevent look-ahead bias.

B. Data Preprocessing and Feature Extraction

During data preprocessing, missing values were addressed using linear interpolation to reduce the impact of data collection inconsistencies. Key features—including maximum, minimum, and average bandwidth (in bps)—were extracted to improve pattern recognition for time-series forecasting. To ensure uniform feature contribution, the data were standardized to have zero mean and unit variance using a scaling technique.

To further enrich the dataset, categorical time-based features like 'Day of the Week', 'Hour of the Day', and 'Non-Working Day' were all converted into numerical form using one-hot encoding.

TABLE III
RESULTS OF EACH SINGLE MODEL

Model	Lag	Avg. MAE	Avg. RMSE
LSTM [13]	8	12.98 Mbps	21.88 Mbps
	16	10.60 Mbps	16.98 Mbps
	24	10.82 Mbps	17.63 Mbps
CNN-GRU [14]	8	8.29 Mbps	12.84 Mbps
	16	8.89 Mbps	13.26 Mbps
	24	8.62 Mbps	13.26 Mbps
CNN-LSTM [15]	8	8.12 Mbps	12.61 Mbps
	16	8.14 Mbps	12.14 Mbps
	24	8.11 Mbps	12.65 Mbps
TCN [16], [25]	8	7.22 Mbps	10.89 Mbps
	16	7.09 Mbps	10.50 Mbps
	24	**6.99 Mbps**	**10.02 Mbps**

C. Configuration of Time Series Forecasting Models

To select the best base predictor (forecasting model) for the NNCT-based weight integration, we compared several leading time series forecasting models, including LSTM, CNN-GRU, CNN-LSTM, and TCN. The detailed configurations are provided, with most settings aligned with those in the relevant literature. All models use the Adaptive Moment Estimation (Adam) optimizer with a learning rate of 0.001.

In our experiments, a batch size of 128 was used, and an early stopping mechanism was implemented to monitor validation performance and prevent overfitting. A relatively high maximum epoch count was set to allow sufficient training when necessary. This setup was designed to strike a balance between thorough model training and effective overfitting control.

D. Case I: Single-Model Performance Analysis

In real-time applications, a sliding window approach is used, where the model inputs only the most recent 8, 16, or 24 hours of data (lags), commonly selected for hour-ahead forecasting. Older data is discarded as new information becomes available. Table III displays the results of each single model. To ensure credibility, both mean absolute error (MAE) and root mean square error (RMSE) are averaged over five runs.

The results indicate that the LSTM model—an RNN variant—initially yielded relatively poor performance. However,

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TABLE IV
RESULTS OF DIFFERENT WEIGHT INTEGRATION METHODS USING TCNS

Method	MAE Average / SD	RMSE Average / SD
Simple Averaging	7.05 / 0.00 Mbps	10.68 / 0.00 Mbps
GA	6.18 / 0.14 Mbps	9.40 / 0.18 Mbps
PSO	6.07 / 0.15 Mbps	9.25 / 0.25 Mbps
L-BFGS-B	5.89 / 0.00 Mbps	8.90 / 0.00 Mbps
L-BFGS-B + NNCT(Prop.)	**5.69 / 0.00 Mbps**	**8.64 / 0.00 Mbps**

integrating CNN substantially improved the accuracy of both CNN-GRU and CNN-LSTM models, underscoring CNN's strength in feature extraction from time-series data. Among them, the CNN-LSTM model achieved a MAE of 8.11 Mbps and an RMSE of 12.65 Mbps.

In addition, TCN model reached a MAE of 6.99 Mbps and an RMSE of 10.02 Mbps, achieving the best performance among the four models. Compared to CNN-LSTM, MAE and RMSE were reduced by 13.8% and 20.8%, respectively. Therefore, this demonstrated the ability of TCN in internet traffic prediction, validating our hypothesis.

E. Case II: Weight Integration and Optimization

To improve accuracy and robustness, weight integration was applied using the optimal model from Case I—the TCN. The model was configured with 10 estimators and a lag (delayed value) of 8. Although this lag did not achieve the highest accuracy, it was chosen because the TCN exhibited minimal error variance across different lag settings. This conservative choice helps prevent overfitting and avoids overly optimistic performance assessments, especially given the difficulty of determining the optimal lag for the test set in advance.

The results of weight integration in Table IV were obtained over five optimization runs. They showed that using only the average weight, accuracy was similar to that of a single TCN. In addition, metaheuristic algorithms like the genetic algorithm (GA) and particle swarm optimization (PSO), as configured in [20], were applied to optimize the weights. These methods notably improved accuracy and reduced error compared to simple averaging and the single TCN by about 12-14% in both error metrics. However, due to the inherent randomness of metaheuristic algorithms, the results showed a high standard deviation (SD), indicating less stable results across multiple runs.

In addition, by introducing L-BFGS-B, the MAE and RMSE decreased to 5.89 Mbps and 8.90 Mbps, respectively, representing reductions of 15.7% and 11.1% compared to the single TCN. Since L-BFGS-B is a deterministic optimization algorithm, it consistently produces the same result, leading to an SD of 0. Moreover, by applying NNCT, the errors decreased further, with the MAE and RMSE reaching 5.69 Mbps and 8.64 Mbps, respectively. These results demonstrate the effectiveness of the proposed method by enhancing robustness through ensemble learning, overcoming the lack of an empirical basis for weights, and offering greater flexibility in weight determination with NNCT. Moreover, L-BFGS-B ensures stable results compared to metaheuristic algorithms, ultimately improving forecasting accuracy over other methods.

V. CONCLUSION AND FUTURE WORKS

In this study, we proposed a novel internet traffic prediction method by integrating TCNs with a new NNCT-based weight integration approach. TCNs enhanced the handling of time-series data by effectively recognizing long-term dependencies, outperforming a CNN-LSTM with a 20.8% reduction in RMSE. The NNCT-based weight integration further enhanced performance by providing increased flexibility and greater optimization potential. Using L-BFGS-B for weight tuning enabled precise optimization, leading to improved accuracy and stability compared to metaheuristic algorithms. Experimental results demonstrated notable performance gains. The success of our approach lay in the effective integration of core technologies and methods, while maintaining a simple and powerful methodology. This paper presented an initial study on applying NNCT-based weight integration for internet traffic prediction. Future work could explore specialized handling of different traffic patterns for further improvement, thereby enhancing network resource management efficiency.

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REFERENCES

[1] M. Boguñá, F. Papadopoulos, and D. Krioukov, "Sustaining the Internet with hyperbolic mapping," *Nat. Commun.*, Art. no. 62, Sep. 2010.

[2] B. Yin, J. Tang and M. Went, "Maximizing the Connectivity of Wireless Network Slicing Enabled Industrial Internet-of-Things," *2021 IEEE GLOBECOM,* Madrid, Spain, 2021, pp. 01-06.

[3] D. Bega, M. Gramaglia, M. Fiore, A. Banchs, and X. Costa-Pérez, "DeepCog: Optimizing Resource Provisioning in Network Slicing With AI-Based Capacity Forecasting," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 2, pp. 361-376, Feb. 2020.

[4] G. Wan, A. Y. Dawod, S. Chanaim, and S. S. Ramasamy, "Hotspots and trends of environmental, social and governance (ESG) research: a bibliometric analysis," *Data Sci. Manag.*, vol. 6, no. 2, pp. 65–75, 2023.

[5] D. H. L. Oliveira, T. P. de Araujo, and R. L. Gomes, "An adaptive forecasting model for slice allocation in softwarized networks," *IEEE Trans. Netw. Serv. Manag.*, vol. 18, no. 1, pp. 94-103, March 2021.

[6] G. Wu, B. Li, J. Niu, D. Gao, and R. Zhang, "Study on service demand model of backbone communication network in new power system," *2023 Int. Conf. Smart Electr. Grid Renew. Energy (SEGRE)*, Changsha, China, 2023, pp. 153-159.

[7] A. Pouttu *et al.*, "6G white paper on validation and trials for verticals towards 2030s," *6G Res. Visions*, vol. 4, 2020.

[8] H. Malik and A. Pouttu, "Integration of LSTM based Model to guide short-term energy forecasting for green ICT networks in smart grids," *2022 IEEE SmartGridComm*, Singapore, 2022, pp. 290-295.

[9] M. Lippi, M. Bertini, and P. Frasconi, "Short-term traffic flow forecasting: An experimental comparison of time-series analysis and supervised learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 14, no. 2, pp. 871-882, June 2013.

[10] G. Li, Y. Li and F. Roozitalab, "Midterm Load Forecasting: A Multistep Approach Based on Phase Space Reconstruction and Support Vector Machine," *IEEE Syst. J.*, vol. 14, no. 4, pp. 4967-4977, Dec. 2020.

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[11] P. Wang *et al.*, "An online electricity market price forecasting method via random forest," *IEEE Trans. Ind. Appl.*, vol. 58, no. 6, pp. 7013-7021, Nov.-Dec. 2022.

[12] S. L. Frenkel and V. N. Zakharov, "Internet traffic prediction model," *Sci. Tech. Inf. Process.*, vol. 5, 2023.

[13] W. Kong *et al.*, "Short-term residential load forecasting based on LSTM recurrent neural network," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 841-851, Jan. 2019.

[14] M. Jayashankara, P. Shah, A. Sharma, P. Chanak, and S. K. Singh, "A novel approach for short-term energy forecasting in smart buildings," *IEEE Sens. J.*, vol. 23, no. 5, pp. 5307-5314, March 1, 2023.

[15] A. Unlu, M. Peña, and Z. Wang, "Comparison of the Combined Deep Learning Methods for Load Forecasting," *Proc. 2023 IEEE PES ISGT*, Washington, DC, USA, 2023.

[16] S. Bai, J. Z. Kolter, and V. Koltun, "An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling," arXiv:1803.01271 [cs.LG], Mar. 2018.

[17] C.-C. Kao, C.-W. Chang, C.-P. Cho, and J.-Y. Shun, "Deep Learning and Ensemble Learning for Traffic Load Prediction in Real Network," *Proc. 2020 IEEE ECICE*, Taiwan, 2020, pp. 36-39.

[18] L. Xiao, J. Wang, Y. Dong, and J. Wu, "Combined forecasting models for wind energy forecasting: A case study in China," *Renewable Sustain. Energy Rev.*, vol. 44, pp. 271–288, Apr. 2015.

[19] T. J. Ikonen, F. Corona, and I. Harjunkoski, "Likelihood maximization of lifetime distributions with bathtub-shaped failure rate," *IEEE Trans. Reliab.*, vol. 72, no. 2, pp. 759-773, June 2023.

[20] F. Zhao, G.-Q. Zeng, and K.-D. Lu, "EnLSTM-WPEO: Short-term traffic flow prediction by ensemble LSTM, NNCT weight integration, and population extremal optimization," *IEEE Trans. Veh. Technol.*, vol. 69, no. 1, pp. 101-113, Jan. 2020.

[21] F. Cassola and M. Burlando, "Wind speed and wind energy forecast through Kalman filtering of numerical weather prediction model output," *Appl. Energy*, vol. 99, pp. 154–166, Nov. 2012.

[22] S. Huang, Y. Sun, and Q. Wu, "Stochastic economic dispatch with wind using versatile probability distribution and L-BFGS-B based dual decomposition," *IEEE Trans. Power Syst.*, vol. 33, no. 6, pp. 6254-6263, Nov. 2018.

[23] Q. Chen, Y. -B. Liu, M. -F. Ge, J. Liu and L. Wang, "A Novel Bayesian-Optimization-Based Adversarial TCN for RUL Prediction of Bearings," *IEEE Sens. J.*, vol. 22, no. 21, pp. 20968-20977, 1 Nov.1, 2022.

[24] R. Gong, J. Li, and C. Wang, "Remaining useful life prediction based on multisensor fusion and attention TCN-BiGRU model," *IEEE Sens. J.*, vol. 22, no. 21, pp. 21101-21110, Nov. 1, 2022.

[25] P. Remy, "Keras Temporal Convolutional Network," GitHub repository, https://github.com/philipperemy/keras-tcn, accessed April 01, 2024.

[26] R. L. Gomes, "Bandwidth usage of university campus," 2020, doi: 10.21227/jw40-y336. *IEEE Dataport*. [Online].

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